DO NOW — RECALL

Search and sort: comparisons and Big O

Name: _____ Class: _____ Date: _____

1. What property does binary search need that linear search does not?

2. What time complexity is typical of merge sort?

3. For n=5, how many times does a single for-loop from 0 to n-1 run?

4. A nested loop runs 4 times outside and 4 times inside. How many inner operations occur?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Sorted input.

2. Model answer

O(n log n).

3. Model answer

5 times.

4. Model answer

16 operations.

Vocabulary — know these terms

time complexity — How an algorithm's work grows with input size.

O(n) — Work proportional to input count.

O(log n) — Work proportional to the number of halvings.

O(n²) — Quadratic growth, often nested loops.

trade-off — A benefit balanced against another cost.

NEW LEARNING

Read, annotate and explain

Count what grows

Big O describes how work increases with input size, not the exact milliseconds on one computer.

Compare search

Linear search can take n checks; binary search on sorted input roughly halves the remaining candidates each step.

Compare sorting

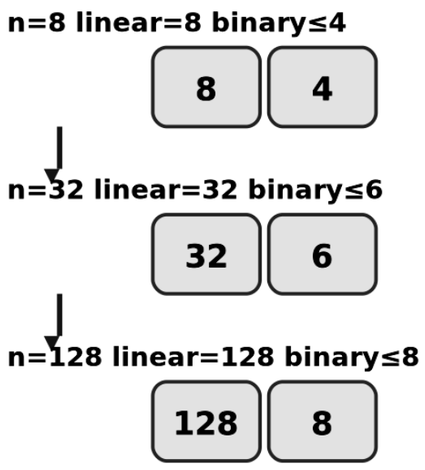
A simple nested comparison loop can be $O(n^2)$, whereas merge sort is $O(n \log n)$.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Read the code; predict its effect



Worked algorithm states: n=8 linear=8 binary≤4 8,4; n=32 linear=32 binary≤6 32,6; n=128 linear=128 binary≤8 128,8

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model solution

Linear worst case = 64 comparisons; binary worst case at most 7. Binary grows much more slowly.

IMPROVE: Explain one correction to your own reasoning.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

O(log n) means execution time is literally log n seconds.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

I DO — WORKED EXAMPLE

Follow the worked method

For 16 values, compare maximum comparisons of linear and binary search.

Predict first

Assume one successful or unsuccessful lookup in sorted data.

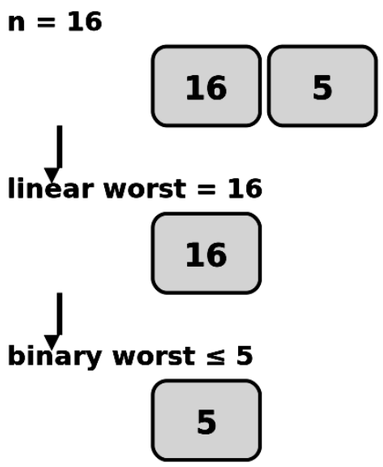
Model solution

Linear search can inspect all 16 elements; binary search checks at most 5 midpoints using this inclusive implementation.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

For 16 values, compare maximum comparisons of linear and binary search.



Worked algorithm states: n = 16 16,5; linear worst = 16 16; binary worst ≤ 5 5

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Is binary search always better than linear search? Give one counterexample.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Name the complexity of a single loop visiting n values.

Question 2

An array has 128 ordered items. Compare linear and binary worst-case checks.

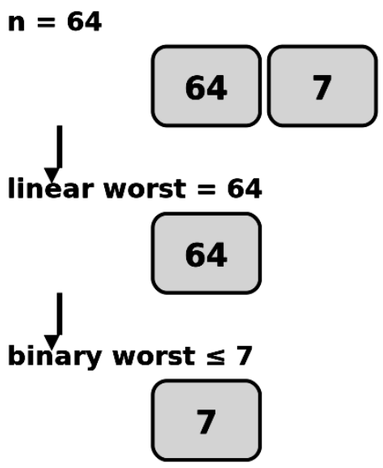
WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Repeat the comparison for 64 ordered values.



Worked algorithm states: n = 64 64,7; linear worst = 64 64; binary worst ≤ 7 7

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first What happens when you double 32 to 64?

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.